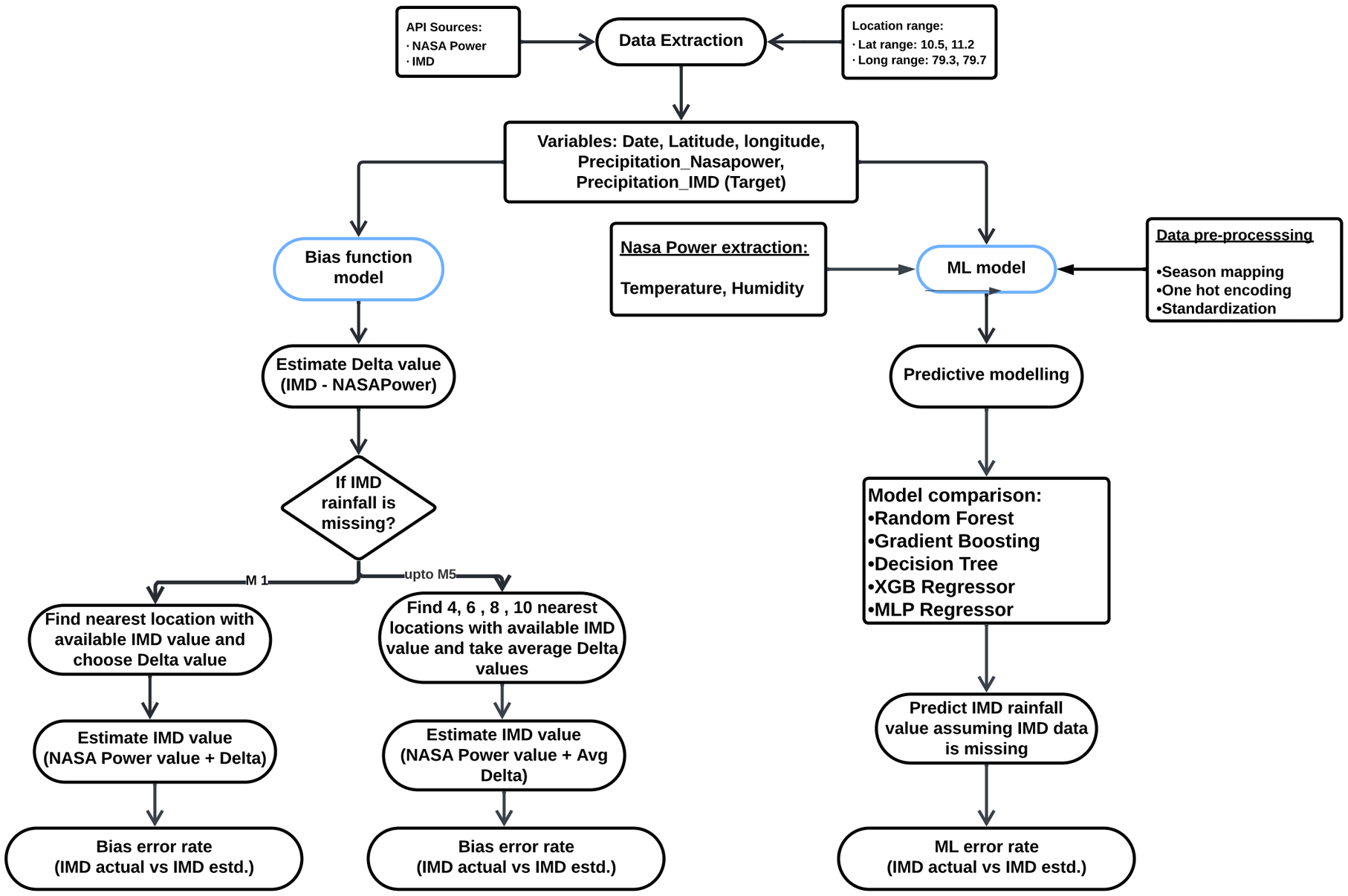
API Sources:
· NASA Power
· IMD
Data Extraction
Location range:
· Lat range: 10.5, 11.2
· Long range: 79.3, 79.7
Variables: Date, Latitude, longitude, Precipitation_Nasapower, Precipitation_IMD (Target)
Bias function model
Nasa Power extraction:
Temperature, Humidity
ML model
Data pre-processsing
•Season mapping
•One hot encoding
•Standardization
Estimate Delta value (IMD - NASAPower)
Predictive modelling
If IMD rainfall is missing?
M 1
upto M5
Model comparison:
•Random Forest
•Gradient Boosting
•Decision Tree
•XGB Regressor
•MLP Regressor
Find nearest location with available IMD value and choose Delta value
Find 4, 6 , 8 , 10 nearest locations with available IMD value and take average Delta values
Predict IMD rainfall value assuming IMD data is missing
Estimate IMD value (NASA Power value + Delta)
Estimate IMD value (NASA Power value + Avg Delta)
Bias error rate (IMD actual vs IMD estd.)
Bias error rate (IMD actual vs IMD estd.)
ML error rate (IMD actual vs IMD estd.)